YONGGANG JIANG

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EDUCATION

Max Planck Institute for Informatics *Apr. 2023 – Present*
PhD in Computer Science
Advisor: Danupon Nanongkai

Max Planck Institute for Informatics *Oct. 2021 – Mar. 2023*
Preparatory Phase for Doctoral Study

Nanjing University *Sep. 2017 – Jul. 2021*
B.S. in Computer Science and Technology
GPA: 94.4/100, Ranking: 1/31

University of California, Berkeley *Jan. 2020 – May 2020*
Berkeley International Study Program
GPA: 4.0/4.0

HONORS & AWARDS

Google PhD Fellowship	*2025*
Distinguished Paper Award, SPAA 2025	*2025*
Outstanding Graduate, Nanjing University	*2021*
National Elite Program Scholarship, Outstanding Prize (top 1)	2018, 2019, 2020
China Collegiate Programming Contest (CCPC), Gold Medal	*2018*
National Olympiad in Informatics in Provinces (NOIP), First Prize	*2015*

RESEARCH INTERESTS

I work broadly at the intersection of algorithms and mathematics in computer science. My current research focuses on algorithm design and analysis for fundamental graph problems. I also work on the theory of parallel and distributed computing.

PUBLICATIONS

* Authors are listed alphabetically with **equal contribution** (see convention in mathematics).

Manuscripts under review

[1] **The Complexity of Distributed Minimum Weight Cycle Approximation**
Yi-Jun Chang, Yanyu Chen, Dipan Dey, Yonggang Jiang, Gopinath Mishra, Hung Thuan Nguyen, Mingyang Yang.

[2] **Approximating Directed Minimum Cut and Arborescence Packing via Directed Expander Hierarchies**
Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Benyu Wang.

[3] **Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth**
Yonggang Jiang, Changki Yun.

Published

[1] **Deterministic Negative-Weight Shortest Paths in Nearly Linear Time via Path Covers**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
ACM Symposium on Theory of Computing (STOC) 2026.

[2] **DAG Projections: Reducing Distance and Flow Problems to DAGs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
ACM Symposium on Theory of Computing (STOC) 2026.

[3] **Reviving Thorup's Shortcut Conjecture**
Aaron Bernstein, Henry Fleischmann, George Z. Li, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
ACM Symposium on Theory of Computing (STOC) 2026.

[4] **Perfect Simulation of Las Vegas Algorithms via Local Computation**
Xinyu Fu, Yonggang Jiang, Yitong Yin.
Innovations in Theoretical Computer Science (ITCS) 2026.

[5] **Reducing Shortcut and Hopset Constructions to Shallow Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[6] **Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs**
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[7] **Minimum s-t Cuts with Fewer Cut Queries**
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[8] **Shortcuts and Transitive-Closure Spanners Approximation**
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[9] **New Oracles and Labeling Schemes for Vertex Cut Queries**
Yonggang Jiang, Merav Parter, Asaf Petruschka.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[10] **Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work**
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.

[11] **Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances**
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award

[12] **Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness**
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[13] **Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations**

Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[14] **Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights**
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.

[15] **Finding a Small Vertex Cut on Distributed Networks**
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.

[16] **Robust and Optimal Contention Resolution without Collision Detection**
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.

[17] **Tight Trade-off in Contention Resolution without Collision Detection**
Haimin Chen, Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Principles of Distributed Computing (PODC) 2021.

Notes

[1] **A Subquadratic Two-Party Communication Protocol for Minimum Cost Flow**
Hossein Gholizadeh, Yonggang Jiang.

RESEARCH VISITS & INVITED TALKS

The University of Hong Kong, Hongkong, China *Dec. 2025*
Hosted by Weiming Feng. Talk: Deterministic Negative-Weight Shortest Paths in Nearly Linear Time via Path Covers
Also at HKUST(GZ), PKU, USTC, NJU

ETH Zurich, Zurich, Switzerland *Nov. 2025*
Hosted by Sorrachai Yingchareonthawornchai. Talk: Deterministic Negative-Weight Shortest Paths in Nearly Linear Time via Path Covers

National University of Singapore, Singapore, *Sep. 2025*
Hosted by Yi-Jun Chang.
Talk: Parallel $(1+\epsilon)$-Approximate Mincost Flow in Almost Optimal Depth and Work

New York University, New York, NY, USA *Sep. 2025*
Hosted by Aaron Bernstein.
Talk: Parallel $(1+\epsilon)$-Approximate Mincost Flow in Almost Optimal Depth and Work

Carnegie Mellon University, Pittsburgh, PA, USA *Aug. 2025*
Hosted by Jason Li.
Talk: Parallel $(1+\epsilon)$-Approximate Mincost Flow in Almost Optimal Depth and Work

University of Michigan, Ann Arbor, MI, USA *Jul. 2025*
Hosted by Thatchaphol Saranurak.

ETH Zurich, Zurich, Switzerland *Feb. 2025*
Hosted by Sorrachai Yingchareonthawornchai.

Tsinghua University, Beijing, China *Dec. 2024*
Hosted by Ran Duan.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Nanjing University, Nanjing, China *Dec. 2024*
Hosted by Yitong Yin.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Shanghai University of Finance and Economics, Shanghai, China *Dec. 2024*
Hosted by Pinyan Lu.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

ETH Zurich, Zurich, Switzerland *Nov. 2024*
Hosted by Weiming Feng.
Talk: New Tools and Faster Algorithms for Vertex Min-Cut

INSAIT, Sofia, Bulgaria *Sep. 2024*
Hosted by Bernhard Haeupler.

University of Warwick, Coventry, UK *Jun. 2024*
Hosted by Sayan Bhattacharya.
Talk: Deterministic k-Vertex Connectivity in k Max-Flows

University of Sheffield, Sheffield, UK *Jun. 2024*
Hosted by Sagnik Mukhopadhyay.

Nanjing University, Nanjing, China *Nov. 2023*
Hosted by Penghui Yao.

Rutgers University, New Brunswick, NJ, USA *Jun. 2023*
Hosted by Aaron Bernstein.

Aalto University, Helsinki, Finland *Apr. 2023*
Hosted by Parinya Chalermsook.

Aalto University, Helsinki, Finland *Dec. 2022*
Hosted by Parinya Chalermsook.

University of Sheffield, Sheffield, UK *May 2022*
Hosted by Sagnik Mukhopadhyay.

CONFERENCE & WORKSHOP TALKS

Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances
SPAA 2025, Portland, OR, USA *Jul. 2025*

Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
STOC 2025, Prague, Czech Republic *Jun. 2025*

Contributed Talk: Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
HALG 2025, Zurich, Switzerland *Jun. 2025*

Contributed Talk: Deterministic k-Vertex Connectivity in k Max-Flows
HALG 2024, Warsaw, Poland *Jun. 2024*

Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights
ESA 2024, London, UK *Jun. 2024*

Finding a Small Vertex Cut on Distributed Networks
STOC 2023, Orlando, FL, USA *Jun. 2023*

Robust and Optimal Contention Resolution without Collision Detection
SPAA 2022, online — *Jul. 2022*

Tight Trade-off in Contention Resolution without Collision Detection
PODC 2021, online — *Jul. 2021*

TEACHING EXPERIENCE

Randomized and Approximation Algorithms, MPI-INF — *Winter 2025*
Role: Teaching Assistant.

Seminar: Foundations of Machine Learning, MPI-INF — *Summer 2023*
Role: Coordinator and Speaker.

Data Structures and Algorithms, Nanjing University — *Fall 2020, Fall 2019*
Role: Teaching Assistant.

STUDENTS & MENTORING

Ioannis Dorkofikis (M.Sc.) — *Spring 2025*
Internship at MPI-INF, from NTUA, Greece.
Now a PhD student at ETH.

Changki Yun (B.Sc.) — *Fall 2024*
Internship at MPI-INF, from SNU, Korea.
Outcome: Paper on parallel vertex connectivity (in submission)

Hossein Gholizadeh (B.Sc.) — *Summer 2024*
Bachelor's Thesis Internship at MPI-INF, from KIT, Germany.
Outcome: Paper in SPAA 2025 (Distinguished Paper Award)

Alireza Danaei (B.Sc.) — *Summer 2023*
Internship at MPI-INF, from Sharif Univ., Iran.
Now a PhD student at MIT.

Vassilis Xanthopoulos (M.Sc.) — *Spring 2023*
Master's Thesis Internship at MPI-INF, from NTUA, Greece.